Image Processing and Medical Imaging

Segmentation

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Segmentation

- One of the most important steps in image processing. Its main goal is to divide an image into parts that have a strong correlation with objects.
- **Segmentation** methods can be divided into three groups according to the dominant feature they employ:
  - **Global knowledge** (for example histogram)
  - **Edge-based** segmentation
  - **Region-based** segmentation

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Edge-based segmentation

- It is based on the edges.
- They can be found by **edge detector**s.
- The most common problems of edge-based segmentation, caused by image noise or unsuitable information in an image, are an edge presence in locations where there is no border, and no edge presence where a real border exists.

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Edge image tresholding

- The small edge values correspond to non-significant gray-level changes resulting from quantization noise, small lighting irregularities, etc.
- Simple tresholding of an edge image can be applied to remove these small values.
- After binarization we can get a so-called **edge image**.

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Border tracing

- First, let us assume that the image with regions is either binary or that regions have been labeled.
- The first goal is to determine inner region borders.
- The inner region border and the outer border have no intersection.

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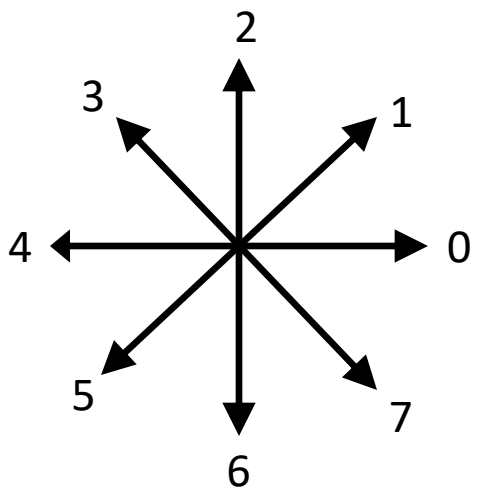
Inner boundary tracing

1. Search the image from top left until a pixel of a new region is found. This pixel P_0 is a starting pixel of the region border. Define a variable *dir* which stores the direction of the previous move along the border from the previous border element to the current border element. Assign *dir=7* if the border is detected in *8*-connectivity.



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Inner boundary tracing (cont.)

2. Search the *3x3* neighborhood of the current pixel in an anti-clokwise direction, beginning the neighborhood search in the pixel positioned in the direction *(dir+7)* mod *8* if *dir* is even, *(dir+6)* mod *8* if dir is odd. The first pixel found with the same value as the current pixel is a new boundary element P_n. Update the *dir* value.

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Inner boundary tracing (cont.)

3. If the current boundary element P_n is equal to the second border element P_1 and if the previous border element P_{n-1} is equal to P_0, stop. Otherwise repeat step 2.

This algorithm is able to find region borders but does not find borders of region holes.

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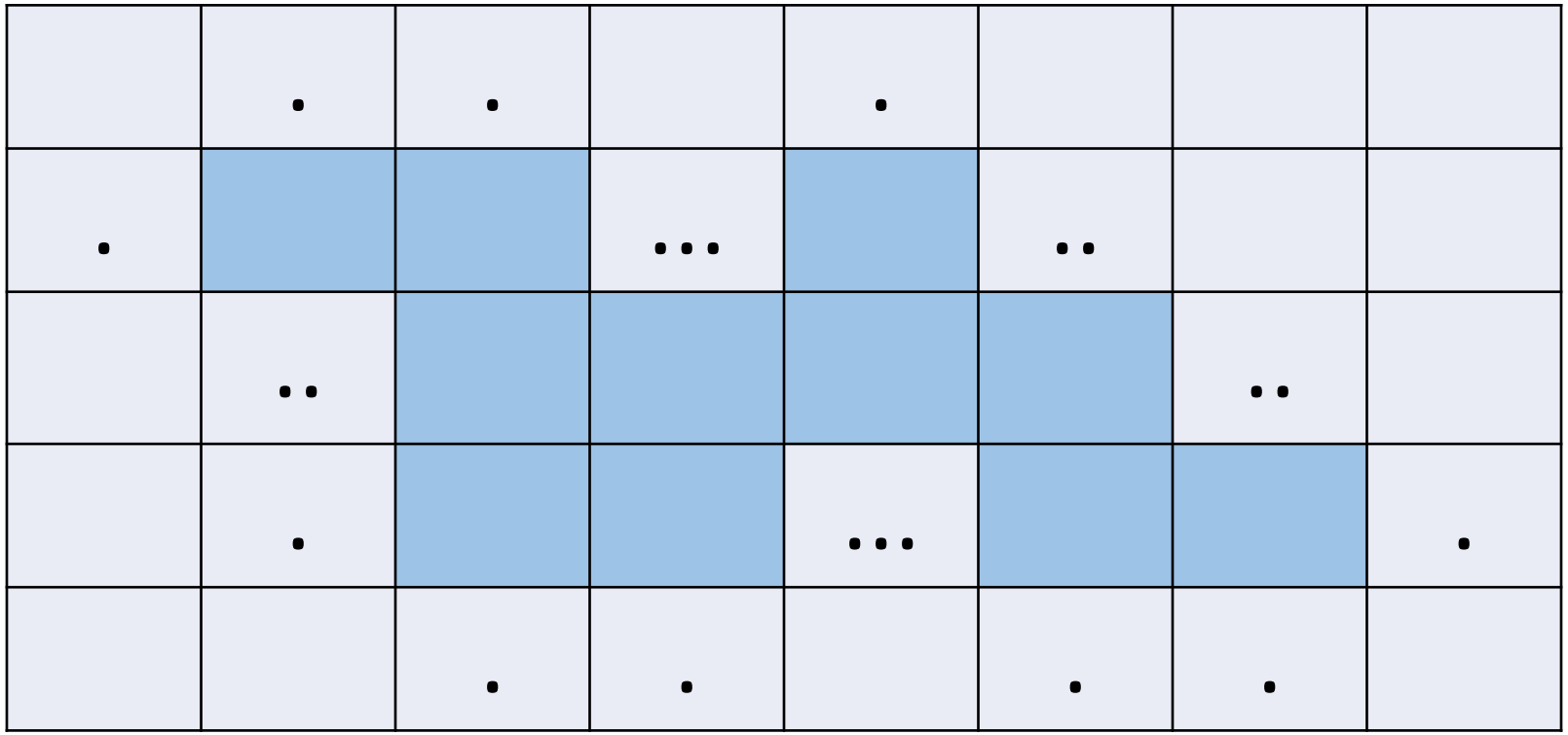
Outer boundary tracing



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Edge detectors

- They are a collection of very important local image pre-processing methods used to locate changes in the intensity functions. Edges are pixels where this function (brightness) changes abruptly.
- An edge is a property attached to an individual pixel and is calculated from the image function behavior in a neighborhood of that pixel.
- The edge magnitude is the magnitude of the gradient, and the edge direction ϕ is rotated with respect to the gradient direction ψ by -90°.

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First differences of the image

$\Delta_i g(I,j)=g(I,j)-g(i-1,j)$

$\Delta_j g(I,j)=g(I,j)-g(I,j-1)$

You can use the sum of the they absolute values.

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Gradient operators

- Gradient operators as a measure of edge sheerness can be divided into three categories:
  - Operators approximating derivates of the image function using differences. Some of them are rotationally invariant (Laplacian), other cases the orientation is estimated on the basis of the best matching of several simple patterns.
  - Operators based on the zero-crossing of the image function second derivative (Canny).
  - Operators which attempt to match an image function to a parametric model of edges.

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Roberts operator

One of the oldest operators (1965). It is very easy to compute as it uses only a 2x2 neighborhood of the current pixel. Its convolution masks are

$$h_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, h_2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

The magnitude of the edges is computed as

$$|g(I,j)-g(i+1,j+1)|+|g(I,j+1)-g(i+1,j)|$$

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Laplace operator

- The Laplace operator ∇^2 is a very popular operator approximating the second derivative which gives the gradient magnitude only.
- The Laplacian operator is approximated in digital images by a convolution sum. A 3x3 mask *h* is often used, for 4-neighborhoods and 8-neighborhoods it is defined as

$$h = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}, h = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

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Prewitt operator

- It approximates the first derivative. The gradient is estimated in eight possible directions, and the convolution result of the greatest magnitude indicates the gradient direction. Larger masks are possible.

$$h_1 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, h_2 = \begin{pmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \end{pmatrix}, h_3 = \begin{pmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{pmatrix}, \ldots$$

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Sobel operator

- The Sobel operator is often used as a simple detector of horizontality and verticality of edges, in which case only masks h_1 and h_3 are used.

$$h_1 = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{pmatrix}, h_2 = \begin{pmatrix} 0 & 1 & 2 \\ -1 & 0 & 1 \\ -2 & -1 & 0 \end{pmatrix}, h_3 = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}, \ldots$$

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Robinson operator

$$h_1 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -2 & 1 \\ -1 & -1 & -1 \end{pmatrix}, h_2 = \begin{pmatrix} 1 & 1 & 1 \\ -1 & -2 & 1 \\ -1 & -1 & 1 \end{pmatrix}, h_3 = \begin{pmatrix} -1 & 1 & 1 \\ -1 & -2 & 1 \\ -1 & 1 & 1 \end{pmatrix}, \ldots$$

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Kirsch operator

$$h_1 = \begin{pmatrix} 3 & 3 & 3 \\ 3 & 0 & 3 \\ -5 & -5 & -5 \end{pmatrix}, h_2 = \begin{pmatrix} 3 & 3 & 3 \\ -5 & 0 & 3 \\ -5 & -5 & 3 \end{pmatrix}, h_3 = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}, \ldots$$

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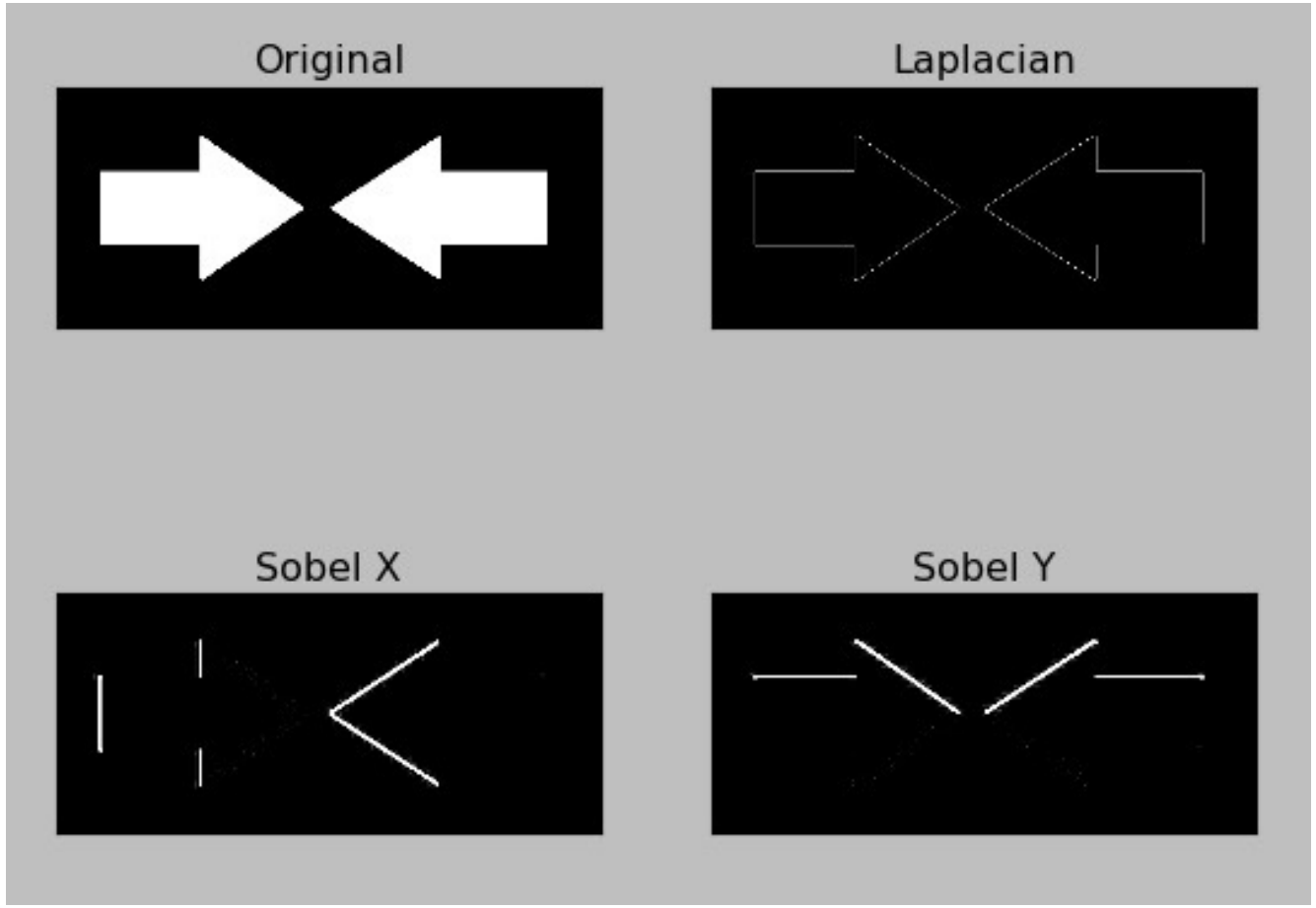
Examples: Sobel, Laplacian



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Source: https://www.meccanismocomplesso.org/en/opencv-python-edge-detection-and-image-gradient-analysis/

Example: Prewitt

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Source: https://www.tutorialspoint.com/dip/prewitt_operator.htm

Zero-crossings of the second derivative

- Our goal is to obtain a second derivative of a smoothed 2D function *f(x,y)*. We have already seen that the Laplace operator ∇^2 gives the second derivative, and is non-directional (isotropic).
- Consider then the Laplacian of an image *f(x,y)* smoothed by a Gaussian. The operation is abbreviated by some authors as LoG, from Laplacian of Gaussian.

$$\nabla^2\left[G(x,y,\sigma) * f(x,y)\right] = \left[\nabla^2 G(x,y,\sigma)\right] * f(x,y)$$

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LoG operator

- Based on the convolution theorem we can get a mask.
- Because of its shape, the inverted LoG operator is commonly called a Mexican hat.

$$\begin{pmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & -2 & -1 & 0 \\ -1 & -2 & 16 & -2 & -1 \\ 0 & -1 & -2 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}$$

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Example: LoG operator

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Source: https://softwarebydefault.com/2013/05/11/image-edge-detection/

Thank you for your attention